

Raman Saparkhan

☎ +974 5037 7425 | rsaparkh@andrew.cmu.edu | [in linkedin.com/in/rsaparkh/](https://www.linkedin.com/in/rsaparkh/) | github.com/roma2023

EDUCATION

Carnegie Mellon University

August 2022 – May 2026

Bachelor of Science in Computer Science, Dean's List

Doha, Qatar

- **Relevant Coursework:** Distributed Systems, Software Engineering, Web Application Development, Generative AI, Algorithms & Design, Deep Learning, Machine Learning, Parallel & Sequential Data Structures & Algorithms.
- **Technical Leadership:** Google Developer Student Club Lead, AWS College Contributor Initiative Team Lead.

EXPERIENCE

Software Engineer Intern

June 2025 - August 2025

UBS *Doha, Qatar*

- Built syntifAI: an AI synthetic-data platform (Python/TypeScript/FastAPI) for 100+ DBs in critical test settings.
- Developed schema-aware parsing to auto-discover tables & generate Profile Maps; cut test-data prep time by 85%+.
- Optimized data processing pipeline via performance tuning on Linux; +35% throughput, 12s/table using Bash scripts.
- Deployed in UBS Doha; led brainstorming sessions to boost team productivity; scalable design for firm-wide usage.

AI Engineer Intern

December 2024 - June 2025

Qatar Computing Research Institute (QCRI)

Doha, Qatar

- Built predictive models integrating LLM & RAG for real-time analytics; used problem solving for data consistency.
- Built a hybrid CoT-PoT ensembling approach; reduced LLM reasoning tasks computational cost by 95% using planning.
- Integrated open-source frameworks (DeepSeekR1, Qwen2.5); cut costs 80% & collaborated on critical model decisions.
- Enabled multi-processing & queuing for concurrent execution; applied critical troubleshooting for workflow speed.

Teaching Assistant for Distributed Systems

August 2024 - November 2024

Carnegie Mellon University, CS Department

Doha, Qatar

- Taught via office hours and 1:1 sessions; developed strong verbal communication skills for computer science topics.
- Applied big data frameworks Ray, MapReduce, Hadoop; used version control systems & software development lifecycle.
- Mentored 35 students on course projects; led brainstorming on system design & performance tuning decision-making.

ML Research Engineer

May 2024 - July 2024

Carnegie Mellon University, Research Department

Doha, Qatar

- Enhanced Pairs Trading by applying NLP into Quant analysis for trading systems; achieved a 15% in annual returns.
- Fine-tuned Llama 3-8B; integrated financial LLMs & lexicons for sentiment analysis; boosted annual returns by 1.5%.
- Built Python development pipeline using scripting language for 300 tickers; retrieved 300k news via Refinitiv API.
- Developed custom backtesting platform for trading algorithm optimization; enhanced software development efficiency.

Software Engineer Contributor

March 2023 - May 2024

Amazon Web Services (AWS) - OpenSearch

Seattle, WA, USA

- Contributed to OpenSearch Python & ML libraries; collaborated with global teams to enhance stability & efficiency.
- Developed snapshot API support; redesigned test splits & integrated new Python development dependency management.
- Featured speaker at OpenSearchCon 2023; shared critical technology insights on open-source contributor development.

PROJECTS

VibePrint | *Node.js, FastAPI, Tailwind CSS, Three.js, WebGL, TripoSR, AWS*

- Developed platform converting images/text/sketches into 3D models with printing services; self-hosted TripoSR AI.
- Implemented real-time 3D rendering with file management; added community features: feed, chat, posts, and sharing.

Lifted AI | *Next.js, FastAPI, LangChain, Supabase, OpenAI*

- Built AI copilot for Project Managers to accelerate task breakdown, allocation & real-time troubleshooting support.
- Implemented embeddings-based risk detection; vector search on Supabase & auto-summarization for meeting reduction.

Find4Cheaper | *Next.js, FastAPI, Supabase, Redis, OpenAI API*

- Built an AI web app aggregating niche marketplace data; comparing prices, availability & shipping trade-offs well.
- Built crawler & real-time search with JSON-LD caching; cut discovery time 90%, indexed 50M+ pages across 500 sites.
- Launched AI shopping assistant enabling natural language product search; automated ranking & de-duplication logic.

FileStack – Distributed File System | *Java, C++, Linux, Java RMI, TCP/IP*

- Built distributed file system (Java/C++) on Linux with TCP/UDP networking, replication & critical fault handling.
- Reduced latency 30% via network routing optimization; used collaborative design & performance tuning methodologies.

TECHNICAL SKILLS

Programming Languages: Python, Bash, C/C++, Java, JavaScript, TypeScript, Dart, Rust, HTML, CSS, SQL, SML

Frameworks & Tools: Git, React, Node.js, Express, PyTorch, TensorFlow, NumPy, Scikit-learn, Pandas, OpenCV

Technologies: UNIX/Linux, Docker, AWS, Azure, Vercel, LangChain, Redis, Firebase, TCP/IP, UDP, SQL Databases